



NETWORK LAYER

Module 3

Services of Network Layer:

- The network layer is responsible for **host-to-host delivery** of datagrams. It **provides services** to the **transport layer** and **receives services** from the **data-link layer**.
- The main services of the network layer are:
 1. Packetizing
 2. Routing
 3. Forwarding
 4. Error Control
 5. Flow Control
 6. Congestion Control
 7. Security

Packetizing:

- Packetizing is the first major duty of the network layer. At the **source**, the network layer **receives data** from the **upper/transport layer**. It **adds a network-layer header** containing **source address, destination address, and other information**. The resulting unit is called a **datagram/packet**.
- At the **destination**, the network layer **removes the header** and delivers the **original payload** to the **upper layer**. If a packet is **fragmented**, the **destination network layer** **reassembles the fragments** before delivering the data.

Routing:

- Routing means **finding the best path** from the **source to the destination**. There can be **multiple possible paths** between **two hosts**. The network layer **determines which path** should be **used**. **Routing protocols** help **routers exchange information** and build their **routing/decision-making tables**.

Forwarding:

- Forwarding is the **actual action performed** by a **router when a packet arrives**. A router **receives a packet** on an **input interface**. It **examines information** in the **packet header**. It **consults its forwarding table**. It selects the **appropriate output interface**. The packet is then sent to **the next network/router**.

Routing	Forwarding
Finds the best path.	Sends packet to the next hop.
Creates/updates routing information.	Uses forwarding table.
Usually performed before packet arrival.	Happens when packet arrives.
Network-wide decision.	Local router decision.

Error Control:

- Error control deals with **detecting/correcting problems** in communication. However, the Internet network layer **does not directly provide error control** for the **whole datagram**.
- **Fragmentation** can **occur at routers**, making **complete error checking** at this layer **inefficient**. IPv4 does have a **checksum for the header**. ICMP can **report certain problems**, such as a **discarded datagram** or **unknown information** in the header.

Flow Control:

- Flow control regulates the **amount of data sent** by a **source** so that **the receiver is not overwhelmed**. Normally, the receiver can **provide feedback** to the sender when it **cannot process data fast enough**.
- However, the Internet network layer **does not directly provide flow control**. Datagrams are sent when the **sender is ready**. Flow control is generally **handled by upper-layer protocols** when required.

Congestion Control:

- Congestion control is a mechanism used to control excessive traffic in a network. Congestion occurs when too many datagrams are present in a part of the network and the number of datagrams sent by source computers exceeds the capacity of the network or routers.
- In such a situation, routers may drop some datagrams. As packets are lost, upper-layer protocols may retransmit them, which can increase the traffic and make congestion worse.
- If congestion continues, the network may reach a point where very few or no datagrams are delivered. In the Internet, congestion control is not directly implemented at the network layer.

Security:

- Security was **not originally built** into the **Internet network layer**. Today, security is important because network communication needs **protection against various threats**.
- **IPSec** is used as a **virtual layer** to **provide security** for the connectionless network-layer service.

DATA LINK LAYER

Module 2

Link Layer Addressing:

- Link-layer addressing is the **addressing mechanism** used at **the data-link layer** to **identify nodes** on a **particular link**. A link-layer address is also called a **physical or MAC address**.
- Since IP addresses alone **cannot identify the links** through which a **packet travels**, link-layer addresses are used for **node-to-node delivery**. **ARP** is used to **map an IP address** to its **corresponding link-layer address**.

Need for Link-Layer Addressing:

- In the Internet, a packet may travel through several networks and routers before reaching its destination. The source and destination IP addresses identify the two end points, but they do not identify the nodes on each individual link. Therefore, another addressing mechanism is required at the data-link layer. A link-layer address is also called a link address, physical address, or MAC address.
- When a datagram is passed from the network layer to the data-link layer, it is encapsulated in a frame. The frame contains two link-layer addresses: a source address and a destination address. These addresses change whenever the frame moves from one link to another.

Three Types of Link-Layer Addresses:

- Some link-layer protocols define **three types of addresses: unicast, multicast, and broadcast.**

Unicast Address:

- A unicast address is **assigned to each host** or to **each interface** of a **router**. It is used for **one-to-one communication**. A frame with a **unicast destination address** is intended for **only one node** on the link.
- In Ethernet, a link-layer address is **48 bits** (6 bytes) and is normally written using **12 hexadecimal digits separated by colons**.

Example: A2:34:45:11:92:F1

Multicast Address:

- A multicast address is used for **one-to-many communication**. A frame is delivered to a **group of nodes**. Its jurisdiction is **local**, meaning it **applies only within the particular link**. Some link-layer protocols support **multicast addresses**.

Broadcast Address:

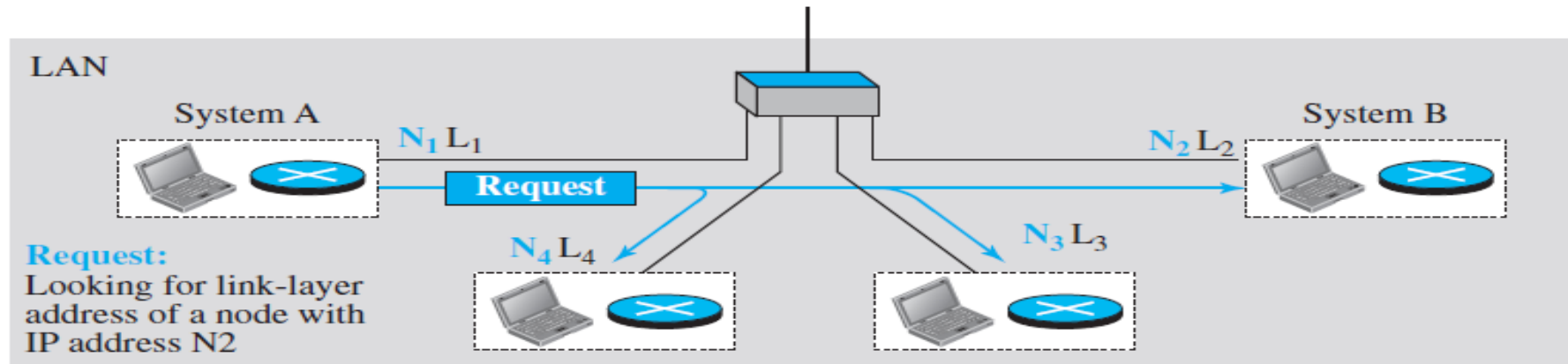
- A broadcast address is used for **one-to-all communication**. A frame with a **broadcast destination address** is sent to **all entities on the link**. In **Ethernet**, the broadcast address consists of **48 ones**, represented as: **FF:FF:FF:FF:FF:FF**.

Address Resolution Protocol (ARP):

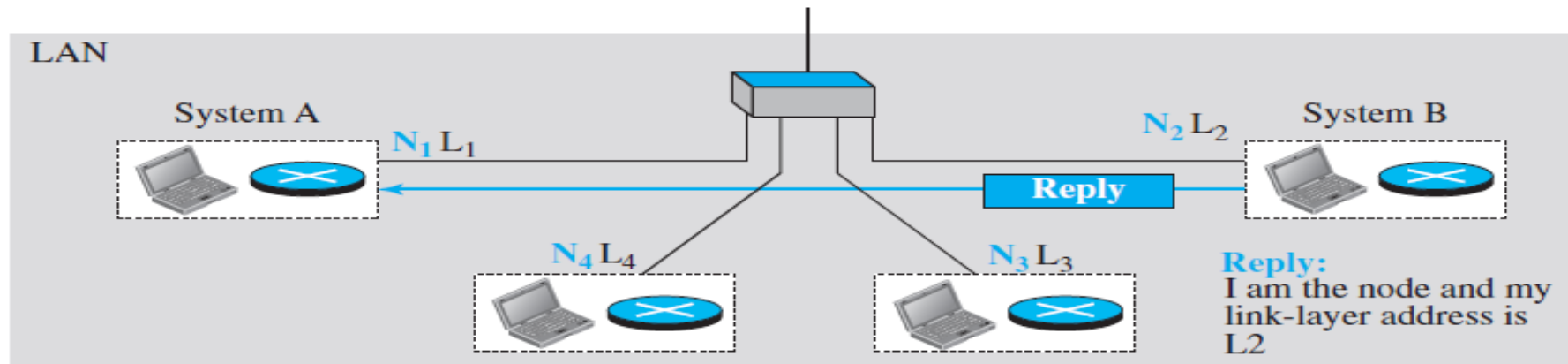
- The network layer knows the IP address of the next node, but the data-link layer needs the link-layer address to deliver the frame. ARP (Address Resolution Protocol) solves this problem.
- ARP accepts an IP address from IP, maps it to the corresponding link-layer address, and passes the link-layer address to the data-link layer.

Working of ARP:

- A source host knows the IP address of the destination/next node. It does not know the corresponding link-layer address.
- It sends an ARP request as a broadcast over the link. All hosts and routers on the link receive the request.
- The node whose IP address matches the requested address sends an ARP reply. The ARP reply contains the node's IP address and link-layer address.
- The reply is sent directly (unicast) to the requesting node. The source can then use the obtained link-layer address to send the frame.



a. ARP request is broadcast



b. ARP reply is unicast

Figure: ARP Operation

- In Figure a, the system on the left (A) has a packet that needs to be delivered to another system (B) with IP address N_2 . System A needs to pass the packet to its data-link layer for the actual delivery, but it does not know the physical address of the recipient. It uses the services of ARP by asking the ARP protocol to send a broadcast ARP request packet to ask for the physical address of a system with an IP address of N_2 .
- This packet is received by every system on the physical network, but only system B will answer it, as shown in Figure b. System B sends an ARP reply packet that includes its physical address. Now system A can send all the packets it has for this destination using the physical address it received.

Interconnection Devices:

- Interconnection devices are used to connect hosts together to form a network or to connect different networks together to form an internetwork.
- The main connecting devices are hubs, link-layer switches, and routers. They operate at different layers of the Internet model.

Hub:

- A hub is a device that **operates only** at the **Physical Layer** (Layer 1). A hub **receives** a **signal** and **regenerates** and **retimes** it.
- It **sends** the **refreshed signal** through **all outgoing ports** except the port from which it **received the signal**. Therefore, a hub essentially **broadcasts the frame** to **all connected stations**.
- A hub has **no filtering capability**. It does **not check MAC/link-layer addresses**. A hub is basically a **multiport repeater**.

- Figure shows that when a packet from station A to station B arrives at the hub, the signal representing the frame is regenerated to remove any possible corrupting noise, but the hub forwards the packet from all outgoing ports except the one from which the signal was received.
- In other words, the frame is broadcast. All stations in the LAN receive the frame, but only station B keeps it. The rest of the stations discard it.

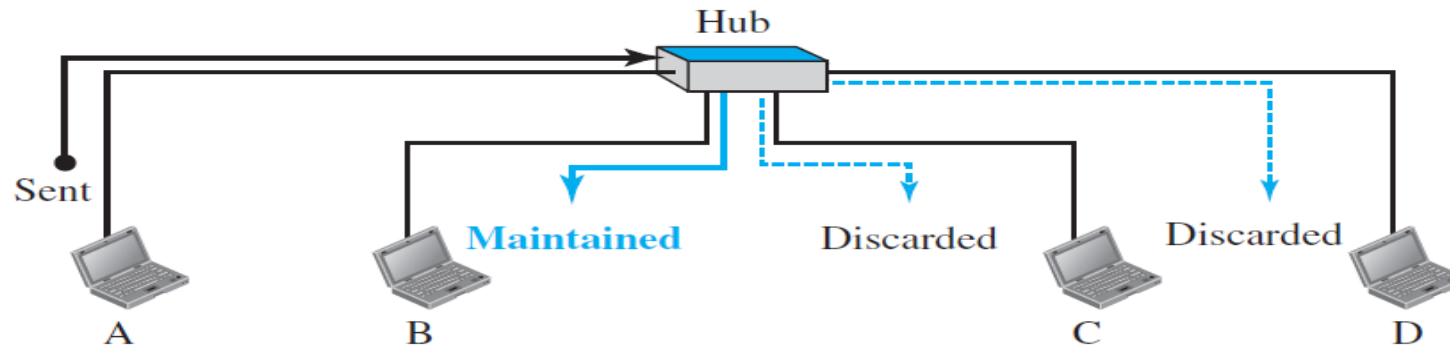


Figure: Hub

Bridge / Link-Layer Switch:

- A bridge connects and divides LANs. A link-layer switch operates at: Physical Layer and Data-Link Layer .
- As a physical-layer device, it regenerates the signal. As a data-link-layer device, it checks the source and destination MAC addresses in the frame.
- A switch has filtering capability. It checks the destination MAC address and decides which outgoing port should receive the frame. Thus, unlike a hub, a switch does not normally send a frame to every port.

- In Figure, we have a LAN with four stations that are connected to a link-layer switch. If a frame destined for station 71:2B:13:45:61:42 arrives at port 1, the link-layer switch consults its table to find the departing port.
- According to its table, frames for 71:2B:13:45:61:42 should be sent out only through port 2; therefore, there is no need for forwarding the frame through other ports.

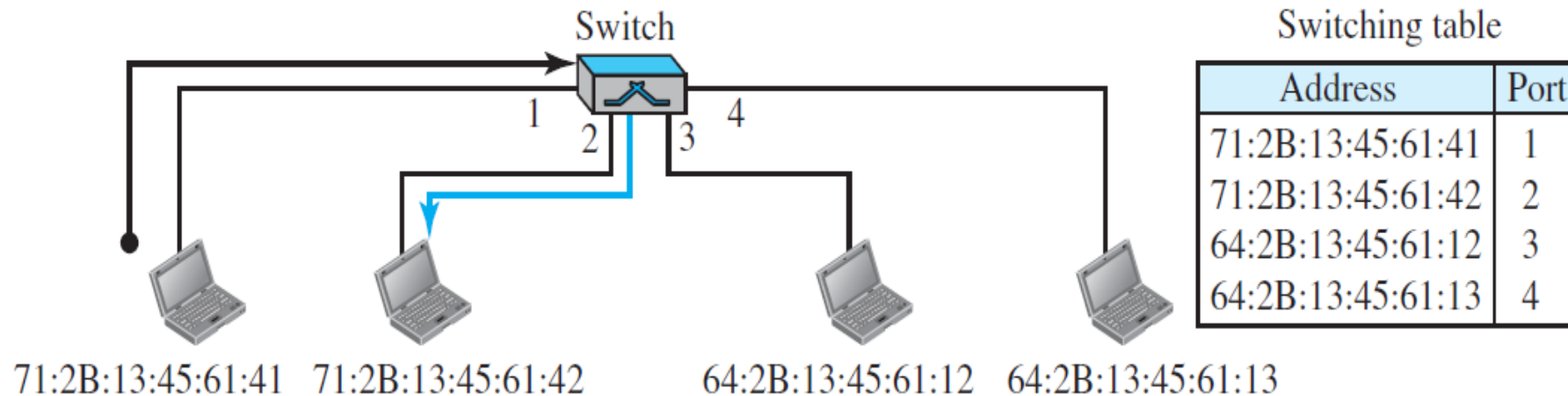


Figure: Switch

Advantages of Switch:

- Eliminates collisions in a switched LAN.
- Increases the average bandwidth available to hosts.
- Can connect devices using different data rates and transmission media, as long as the frame format remains compatible.

Router:

- A router is a **three-layer device**. It operates at: **Physical Layer, Data-Link Layer, and Network Layer**. At the **physical layer**, it **regenerates signals**. At the **data-link layer**, it **checks physical/MAC addresses**. At the **network layer**, it **checks network-layer addresses** such as **IP addresses**.

Functions of Router:

1. Connects **different networks**.
2. Forms an **internetwork** from **independent networks**.
3. Uses **IP/network-layer addresses** to **forward packets**.
4. Each router interface has a **physical address** and **logical (IP) address**.
5. When **forwarding a packet**, the router **changes** the link-layer **source** and **destination addresses** for the next link.

- In Figure, assume an organization has **two separate buildings** with a **Gigabit Ethernet LAN** installed in **each building**. The organization **uses switches** in each LAN. The **two LANs** can be **connected to form** a larger LAN using 10 Gigabit Ethernet **technology** that **speeds up** the **connection** to the **Ethernet** and **the connection** to the **organization server**. A router then can **connect the whole system** to the **Internet**.

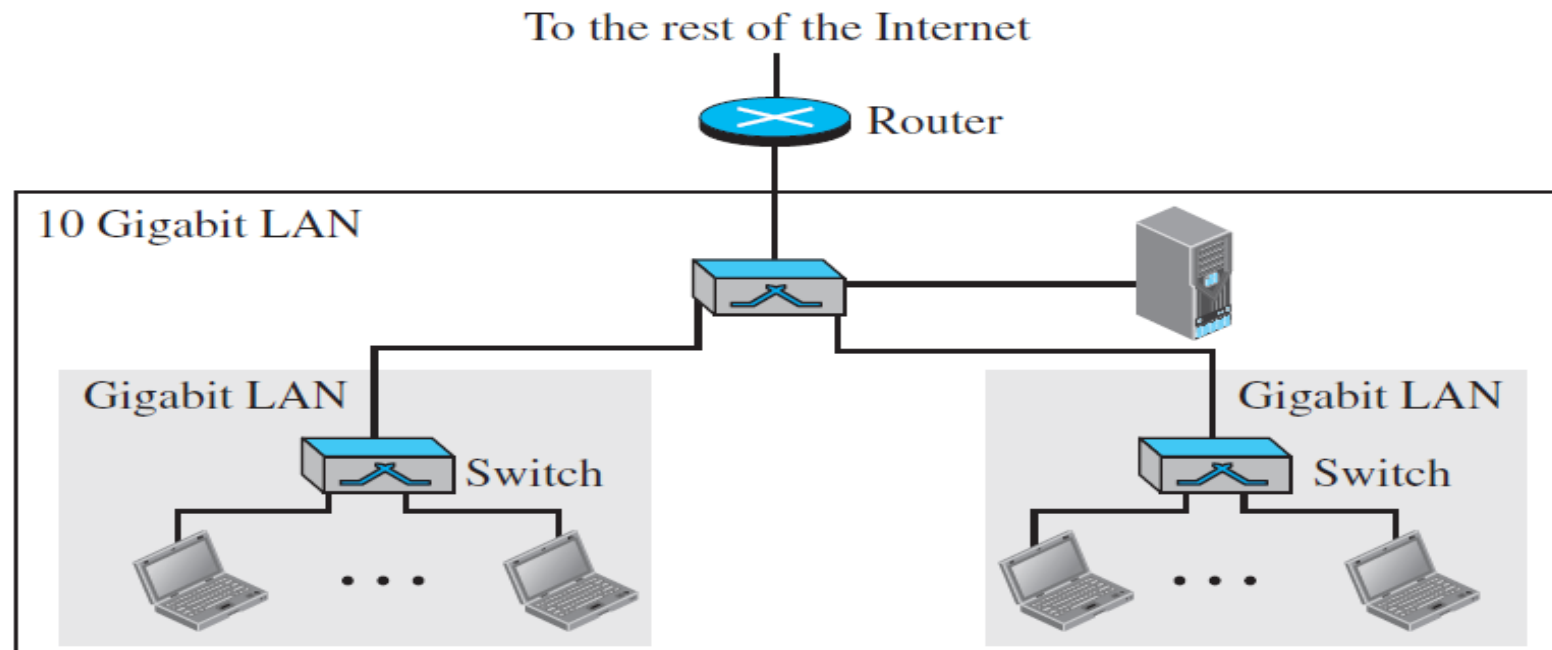


Figure: Router

Difference Between Hub, Bridge/Switch, and Router:

Features	Hub	Switch/Bridge	Router
Layer	Physical Layer (Layer 1).	Physical + Data-Link Layers (Layers 1 & 2).	Physical + Data-Link + Network Layers (Layers 1, 2 & 3).
Main function	Regenerates and forwards signals.	Filters and forwards frames.	Connects different networks and forwards packets.
Address checked	No address checking.	MAC address.	IP address.
Forwarding	Sends signal to all outgoing ports.	Sends frame only to the appropriate port.	Sends packet toward the appropriate network.
Filtering	No filtering.	Has filtering capability.	Uses routing information.
Network connection	Connects devices within a LAN.	Connects/segments LANs.	Connects different networks.
Data unit	Signal/Bits.	Frame.	Packet.
Address of device	No link-layer address.	Uses MAC addresses.	Each interface has MAC and IP addresses.

References:

Data Communications and Networking by Behrouz A. Forouzan.



THANK YOU